Day 41–42: Searching & Sorting

Learn:

- **Linear Search**
- **Binary Search**
- **Bubble Sort**
- **Selection Sort**
- **Insertion Sort**

LINEAR SEARCH

```
public class Main
{
    public static void main(String[] args) {
        int arr[]={1,5,3,8,4};
        int n=arr.length;
        int target=5;
        boolean found=false;
        for(int i=0;i<n;i++){
            if(arr[i]==target){
                found=true;
                break;
            }
        }
            if(found==true){
                System.out.println("found");
            }
            else{
                System.out.println("not found");
            }
```

```
}
}
```

BINARY SEARCH

```
public class Main
{
        public static void main(String[] args) {
            int arr[]={1,3,5,7,9};
            int n=arr.length-1;
            int target=5;
            boolean found=false;
            int low=0,high=arr.length;
            while(low<=high){
                int mid=(high+low)/2;

                if(arr[mid]==target){
                    System.out.println("found");
                    found=true;
                    break;
                }
                else if(target<arr[mid]){
                    high=mid-1;
                }else{
                    low=mid+1;

                }
}
}
```

```
}
```

BUBBLE SORT

```
public class Main
{
	public static void main(String[] args) {
	    int arr[]={1,5,3,8,4};
	    int n=arr.length;
	    for(int i=0;i<n-1;i++){
	        for(int j=0;j<n-1-i;j++){
	            if(arr[j]>arr[j+1]){
	                int temp=arr[j];
	                arr[j]=arr[j+1];
	                arr[j+1]=temp;
	            }
	        }
	    }
	    for(int num :arr){
	        System.out.println(num+" ");
	}
}
}
```

SELECTION SORT

```
public class Main
{
	public static void main(String[] args) {
	    int a[]={1,4,2,8,5,9};
	    for(int i=0;i<a.length-1;i++){
	        int minindex=i;
```

```
            for(int j=i+1;j<a.length;j++){
                if(a[j]<a[minindex]){
                    minindex=j;
                }
            }
                int temp=a[i];
                a[i]=a[minindex];
                a[minindex]=temp;
            }

        for(int num:a){
            System.out.println(num+" ");
        }
}
}
```

- **Insertion Sort**

```
public class Main
{
        public static void main(String[] args) {
            int arr[]={1,7,4,9,5};
            for(int i=1;i<arr.length;i++)
            int key=arr[i];
            int j=i-1;
            while(j>=0&&arr[j]>key){
                arr[j+1]=arr[j];
                j--;
            }
```

```
            arr[j + 1] = key;
        }

        for (int num : arr) {
            System.out.print(num + " ");
        }
    }
}
```